

Patient Readmission Risk Initiative

Business Summary | Data Readiness Update | 2 September 2026

Executive Summary

Over 1.1 million patient records were processed to build a foundation for predicting 30-day hospital readmissions. After cleaning and enriching the data, we retained a high-quality analytical base of 972,551 patients ready for predictive modeling. Roughly 1 in 7 patients (15%) in this population was readmitted within 30 days - representing a significant opportunity for early intervention and cost avoidance.

Key Numbers at a Glance

1,155,000	972,551	15.0%	7
Patient records analyzed	Clean, analysis-ready patient records	Patients readmitted within 30 days	New clinical risk indicators created

What We Did (Steps 1-3)

- **Step 1 - Explored the raw data:** reviewed 1.1M patient records, uncovering data entry errors (e.g., invalid ages), duplicate/inconsistent fields, and a significant imbalance between readmitted and non-readmitted patients.
- **Step 2 - Cleaned the data:** removed unusable and duplicate records, corrected invalid entries, filled in gaps using statistically sound methods, and standardized inconsistent labels (e.g., gender values).
- **Step 3 - Engineered new risk signals:** added 7 clinically meaningful indicators - kidney function stage, liver risk, polypharmacy risk, BMI category, age group, elderly high-dose flag, and a liver-disease ratio - to make the data more predictive and clinically interpretable.

Why This Matters

- Every percentage-point reduction in 30-day readmissions translates directly into lower penalty exposure and improved patient outcomes.
- New risk flags (polypharmacy, kidney/liver risk, elderly high-dose) give care teams early, explainable warning signs - not just a black-box score.
- A clean, de-duplicated dataset of ~972K patients means predictive models trained next will be more accurate and trustworthy for clinical decision-making.

Data Quality Watch-Outs

- Gender field still contains inconsistent entries in source systems; recommend a data-entry standardization fix at the point of capture.
- Drug name variety is very high, which expanded the dataset width; we recommend grouping drugs into therapeutic classes before final modeling for better performance and interpretability.

Next Step: Build & validate the readmission prediction model (Step 4)